

How to Use MITRE ATT&CK with Wazuh SIEM for Threat Hunting

1- Definition

MITRE ATT&CK is:

A global open knowledge base describes tactics and techniques and how it operates in real-world environments.

A specific method the defender can use to identify, detect and response to attack patterns

A specific method the defender can use to map attacker steps like initial access, execution, persistence, defense, evasion, etc.

MITRE ATT&CK is a knowledge base of three things (TTPs):

- **Tactics**, means, what was the attacker's purpose? Why did they penetrate the system?
- **Techniques** means, what methods is, the hacker, using to achieve their purpose, like guessing passwords?
- **Procedures** are the specific details about that attack.

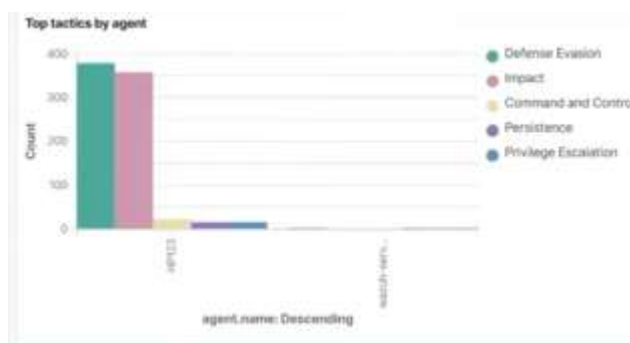
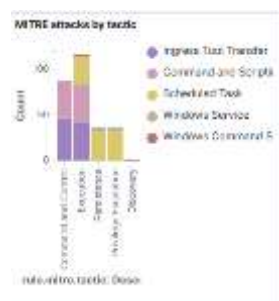
2- The role of MITRE ATT&CK in WAZUH

WAZUH comes with MITRE ATT&CK out of the box. This means that when the agents we have configured, like Windows or Linux agents, send logs, WAZUH analyzes them and informs us that this is not an error, or it was a brute force attack, or its purpose was credential access.

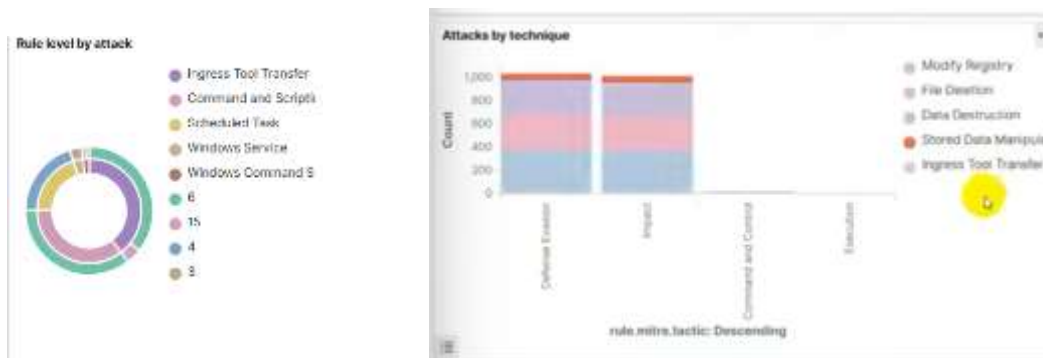
A- How to see it on wazuh?

Click on the menu → Here you will find Threat Intelligence → Click on MITRE ATT&CK, there we can see Tactics and Techniques meaning.

Tactics refer to which attack phase is ongoing, for example, persistence, privilege escalation, command and control, impact, and similar.



Technique means which method is being used, like data destruction, stored data manipulation, brute force attack, etc.



B- Simulation of attack

Go to our Windows agent and run this command in PowerShell:

```
powershell.exe -nop -w hidden -c "Write-Host 'Testing Fileless Detection'"
```

NB: The role of this command is to simulate a benign, "fileless" PowerShell attack pattern to test whether endpoint monitoring tools (like Wazuh SIEM, Sysmon, or EDR solutions) can detect hidden execution behaviors.

Command Switch Breakdown

Each parameter in this command mimics stealth tactics commonly used by threat actors:

- **-nop (-NoProfile):** Prevents PowerShell from loading the user's custom environment profile. Attackers use this to speed up execution, avoid profile-based errors, and bypass customized security scripts.
- **-w hidden (-WindowStyle Hidden):** Hides the PowerShell console window from the user interface. On the target desktop, no command prompt pops up, allowing the process to run silently in the background (**Defense Evasion**).
- **-c (-Command):** Directs PowerShell to execute the inline string immediately following the flag and then exit.
- **"Write-Host 'Testing Fileless Detection'"**: The actual payload being executed. Instead of downloading malware or executing a malicious binary on disk, it simply prints text to the hidden buffer.

This command simply tells PowerShell to print text to the console screen and then exit. It does not:

- Download any files from the internet.
- Modify system settings or the Windows Registry.
- Install software or malware.
- Delete or alter any files on your hard drive.

Let's check this on WAZUH. So, we have:

The command: **Executable file dropped in folder commonly used by malware,**

By checking it, it will tell us which MITRE technique ID has been used (rule.mitre.id = T1105), what tactic is been used (rule.mitre.tactics = command and control) and what technique is been used (rule.mitre.technique = ingress Tool Transfer),

In another way:

Go to menu → threat intelligence → MITRE ATT&CK → go to events, it shows us in detail which tactic was used (**rule.mitre.tactic**) what was its description (**rule.description**) what was the technique (**rule.mitre.id**) and out of all the attacks it is showing us it is displaying the respective tactics and techniques.

Now we copy the technique ID and after that we go to MITRE, we go to search paste it there and check it so will this tell us in detail what the actual purpose of this attack is what are the steps involved



WAZUH
Open Source Security
Monitoring Platform

How to Use MITRE ATT&CK with Wazuh SIEM for Threat Hunting

Map Attacks • Detect • Investigate • Respond



**MITRE
ATT&CK®**
Know the Adversary.
Stop the Attack.

1 What is MITRE ATT&CK?

MITRE ATT&CK is:

- ✓ A global open knowledge base that describes tactics, techniques and how attackers operate in real-world environments.
- ✓ A method for defenders to identify, detect and respond to attack patterns.
- ✓ A way to map attacker steps (initial access, execution, persistence, defense evasion, etc.).

The 3 key components (TTPs):

- Tactics**
What is the attacker's purpose?
• Why did they penetrate the system?
- Techniques**
What methods are used?
(e.g., guessing passwords)
- Procedures**
The specific details about the attack.

MITRE ATT&CK Framework



2 The Role of MITRE ATT&CK in Wazuh

Wazuh comes with MITRE ATT&CK out of the box. When agents (Windows or Linux) send logs, Wazuh analyzes them and tells us if it's an error, a brute force attack, or a credential access attempt.



A- How to see it on Wazuh?

- Go to the menu → Threat Intelligence → MITRE ATT&CK.
- There you can see Tactics and Techniques meaning.

wazuh

- Discover
- Agents
- Threat Intelligence
- MITRE ATT&CK
- IOC Search
- Vulnerability Detection

MITRE ATT&CK

Tactics	Techniques
> Initial Access	10
> Execution	8
> Persistence	7
> Privilege Escalation	6
> ...	...

Tactics refer to which attack phase is ongoing (e.g., persistence, privilege escalation, C2, impact).

Techniques refer to the method being used (e.g., data destruction, brute force, etc.).

3 Simulation of Attack

On your Windows agent, run this command in PowerShell:

```
powershell.exe -nop -w hidden -c "Write-Host 'Testing Fileless Detection'"
```

NB: The role of this command is to simulate a benign, "fileless" PowerShell attack pattern to test whether endpoint monitoring tools (Wazuh, Sysmon, or EDR) can detect hidden execution behaviors.

Command Switch Breakdown

- nop** (-NoProfile) : Prevents loading user profile (speeds up execution, avoids custom scripts).
 - w hidden** (-WindowStyle Hidden) : Hides the PowerShell window (defense evasion).
 - c** (-Command) : Executes the inline command.
- "Write-Host 'Testing Fileless Detection'" : Prints text to the console (no file, no registry change, no download).

This command does NOT:

- Download any files from the internet.
- Modify system settings or the Windows Registry.
- Install software or malware.
- Delete or alter any files on your hard drive.



4 Check the Logs on Wazuh

Go back to the Wazuh dashboard and check the alerts.

wazuh

- Dashboard
- Security Events
- Threat Intelligence
- Agents
- Settings

Security Events

Search...

rule.groups - sysmon

Time	Agent	Rule ID	Description
Sep 13, 2026 15:24	DESKTOP-WIN	100100	Sysmon - Process
Sep 13, 2026 15:24	DESKTOP-WIN	100101	PowerShell detected
Sep 13, 2026 15:25	agent-linux	100200	Auditd - Command

✓ You can filter by rule groups, agent, or provider name (e.g., data.win.system.provider.Name) to confirm that Sysmon logs are actively collecting.

5 Prepare the Environment

A Windows: Add Sysmon & PowerShell logs

In Wazuh (Windows group → agent.conf), add:

```
<!-- Enable Sysmon Logs -->
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>

<!-- Enable PowerShell Logs -->
<localfile>
  <location>Microsoft-Windows-PowerShell/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

B Linux: Enable Auditd logs

In Wazuh (Linux group → agent.conf), add:

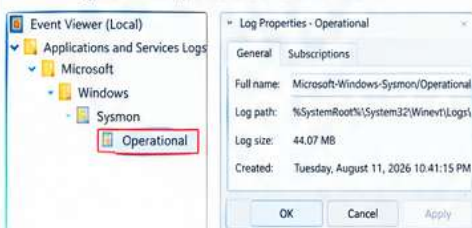
```
<!-- Enable Auditd Logs -->
<localfile>
  <log_format>auditd</log_format>
  <location>/var/log/audit/audit.log</location>
</localfile>

<log_format>syslog</log_format>
<location>/var/log/auth.log</location>
</localfile>
```

6 Verify the Logs

Windows Agent:

- Go to Events Viewer → Applications and Services Logs → Microsoft → Windows → Sysmon → Operational.
- Right-click → Properties to see the channel name.



Linux Agent:

- Enable auditd:


```
sudo apt-get install auditd auditd-plugins -y
```
- Start and enable:


```
sudo systemctl enable auditd
```
- Add config and restart Wazuh.

✓ Check logs on both agents and filter by rule groups to confirm successful collection.

7 Result

✓ Successfully viewing and filtering logs from both multi-platform agents confirms that your SIEM telemetry pipeline is fully operational for threat hunting.



WAZUH



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"Knowledge
Builds a Safer"